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Hades: A Hardware-Resident Substrate for Energy Admission in Intermittent Computing

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The hidden cost of tiny sensors

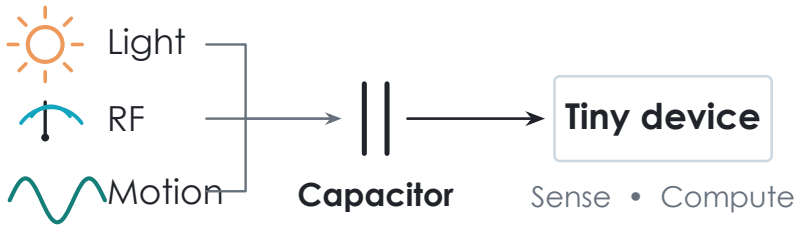
IoT scales. So does battery maintenance.



Battery replacements add maintenance, e-waste, and carbon footprint at IoT scale.

What if sensors harvested energy?

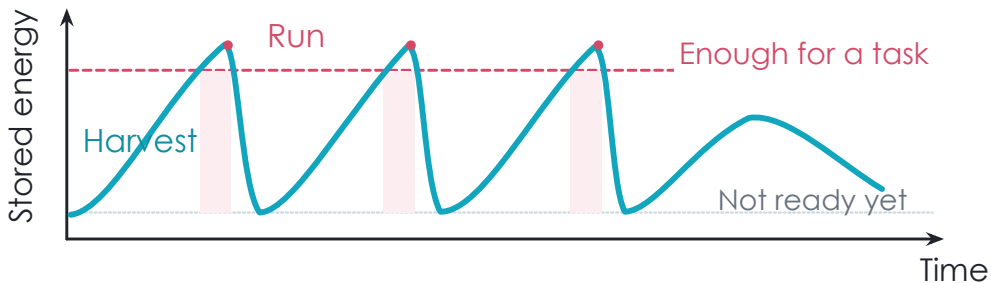
Light, radio waves, and motion can power tiny devices.



The catch: energy arrives in unpredictable bursts.

Energy arrives. Tasks wait.

A capacitor charges. A task runs. Then it waits.



Who decides when the next task can safely start?

Related work: different goals

Prior work improves different parts of batteryless operation.

Software-mediated admission
Mementos • Dewdrop
Runtime and scheduling

Hardware-assisted admission
Failure Sentinels • CapDYN
Monitoring and buffering

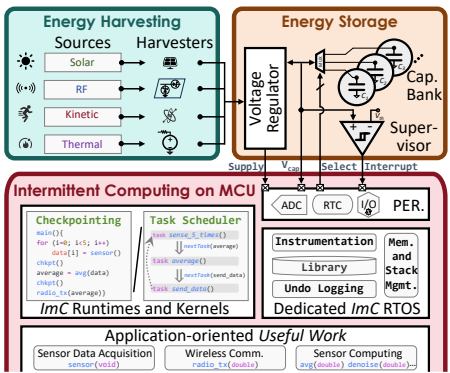
Post-admission execution
Clank • ReplayCache
Persistence and recovery

Pre-admission energy intake
AmbiMax • InfiniteEn • MPPT
Harvesting and buffering

Let us zoom in on task admission.

How does a task get admitted?

A voltage supervisor wakes the microcontroller unit (MCU).



1. Harvest and store
Capacitor banks collect energy.

2. Hardware wakeup
Supervisor wakes the MCU.

3. Software admission

Sample storage via ADC

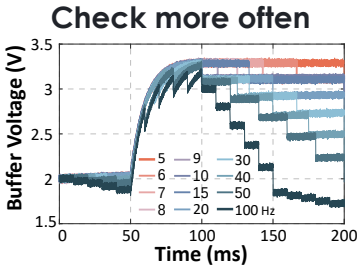
Evaluate available energy

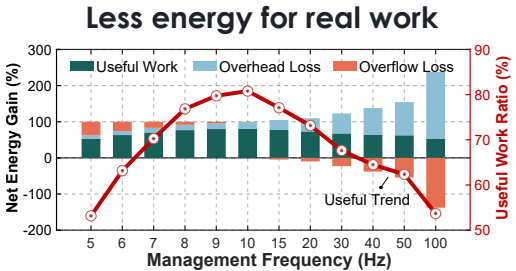
Choose the next task

Execute or return to sleep

Checking energy costs energy

Each wakeup runs ADC sensing and admission code, even when no task can start.

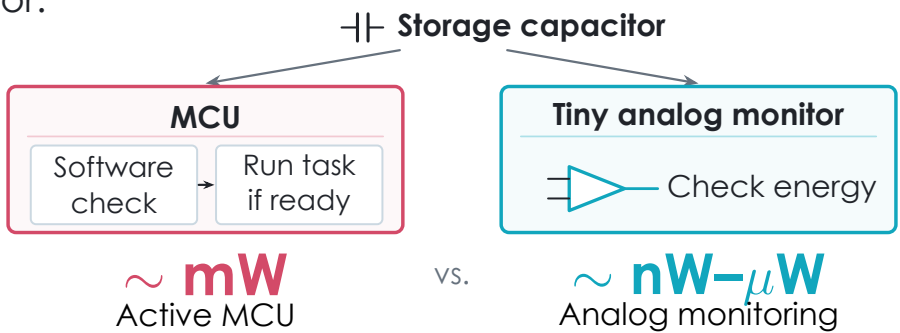




More checking means less energy left for useful tasks.

The processor is power-hungry

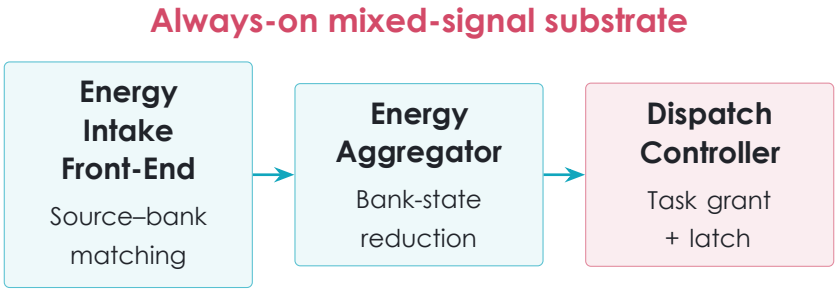
An active MCU uses far more power than a tiny analog monitor.



Why wake an mW-class MCU just to check energy?

Meet Hades: harvest more, spend less

Hades brings more energy into storage and admits tasks while the MCU stays off.

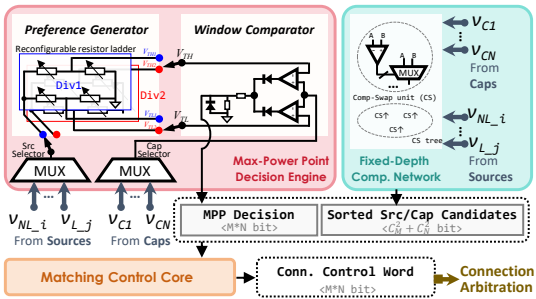


More energy in. Less MCU energy spent on management.

Near-MPPT, without costly tracking

Digital MPPT repeatedly senses and retunes the load.

At μW inputs, the control loop can cost too much.

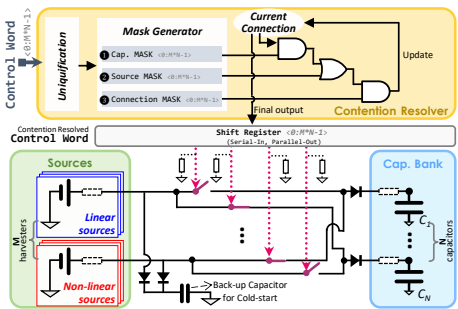


Insight: Each source has a productive voltage window.

Analog comparisons match it to an existing bank voltage.

Compare voltages. Connect safely.

Analog windows identify feasible pairs; arbitration selects a valid connection.



1. Compare
Check voltage windows

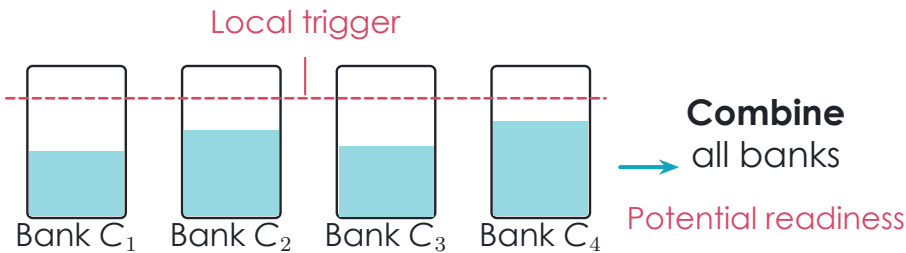
2. Resolve
Avoid source–bank conflicts

3. Commit
Update the switch paths

No continuous ADC-driven load search at runtime.

One bank looks low. Together, enough.

A local voltage trigger can miss energy spread across several banks.



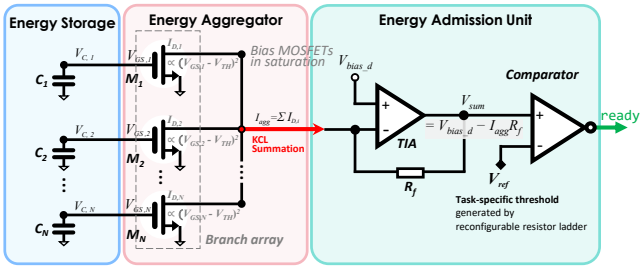
Insight: use the combined reserve, not a single bank voltage.

Let the circuit compute energy

Many banks. One admission decision.

Multiple ADCs High power
Parallel sampling + digital summation

Comparator ladders Coarse
Quantized bank-state estimates



OUR INSIGHT

MOS saturation
Square-law $V \rightarrow I$

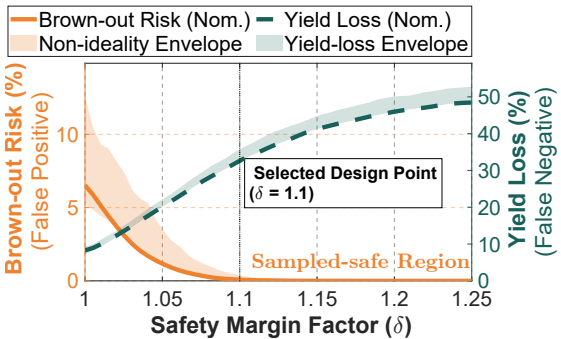
Bank weighting
$K_i \propto C_i$

KCL summation
Parallel current sum

$$I_{agg} = \sum_i I_{D,i} \longrightarrow V_{sum} \xrightarrow{V_{ref,t}} \texttt{ready}$$

A fast comparison needs a safe threshold

The analog sum is a proxy for usable energy. Task profiling and calibration turn it into a conservative start decision.



At the edge

Profile each bounded task

On the device

Load its calibrated $V_{\mathrm{ref},t}$

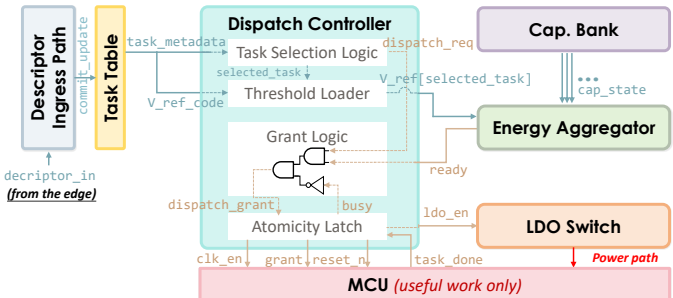
Margin $\delta = 1.1$

0.38%

stressed false positives (95th percentile)

Different tasks, different energy needs

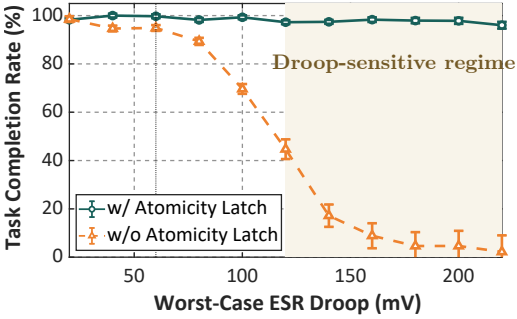
A task's energy demand is profiled ahead of time and stored as a threshold; dispatch needs no MCU-side calculation.



Select task ⟶ load $V_{ref,t}$ ⟶ grant ⟶ release MCU

Once it starts, keep the window open

MCU execution drains the capacitor, so a live voltage comparator may change state before the task finishes.



Latch the grant

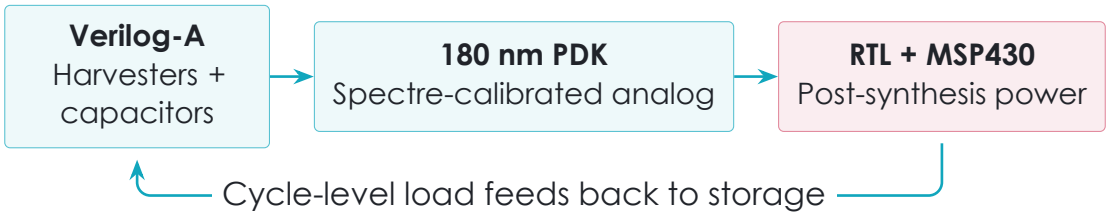
Ignore expected voltage droop

Block overlapping dispatch

Close on task

How we evaluate Hades

We model the harvested energy, hardware circuits, and MCU activity together.



12 embedded workloads
AES, FFT, SHA, etc.

4 energy sources
Solar, RF, thermal, motion

180 nm PDK
PDK-calibrated models

Always on, with little overhead

The admission circuit stays awake so the expensive MCU does not have to.

0.29%	542 nW	13.13 pJ	1.05 μs
evaluated SoC area	always-on power	dynamic energy / decision	decision response

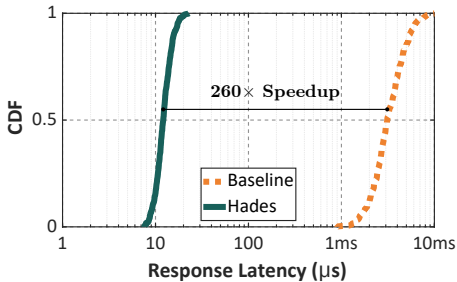
Software: **2.51** μ**J** per attempt $\longrightarrow$ >98% lower modeled admission energy

Post-synthesis digital estimates + transistor-calibrated analog modeling; comparison is not measured silicon power.

Act fast on weak energy

React faster and operate at lower input power.

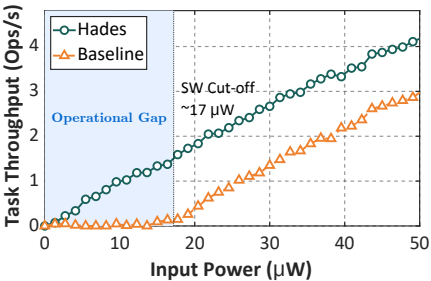
Burst-to-execution latency



12 μs vs. 3.2 ms

Median $\approx 260\times$ faster

Minimum viable input power

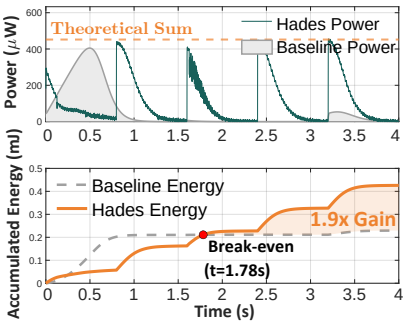


$\sim 2\ \mu W$ vs. $\sim 17\ \mu W$

Under evaluated workload/trace settings

Better matching. More energy stored.

When a source sees a better load, more of its energy reaches storage.



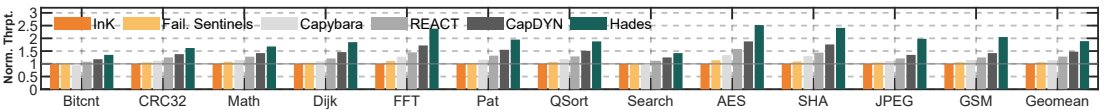
1.9×

energy delivered to storage

Solar + RF matching experiment, 4-second trace

More work gets done

Across 12 workloads, Hades completes more tasks in the same time.



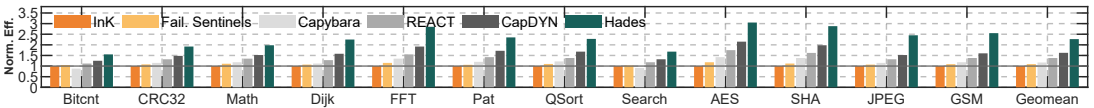
2.2×
geometric-mean vs. InK

1.35×
geometric-mean vs. CapDYN

2.5×
up to vs. InK

More tasks from every joule

With less management overhead, more harvested energy becomes useful work.



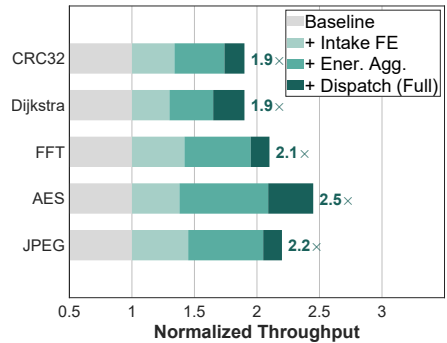
2.4× geometric-mean tasks/J over InK

Why it helps

Collect more
Check less
Dispatch wisely

Why all three blocks matter

Each block adds value. Together, the gains build up.



Front-End
$1.38\times$

+ Aggregator
$1.9\times$

+ Dispatch
up to $2.45\times$

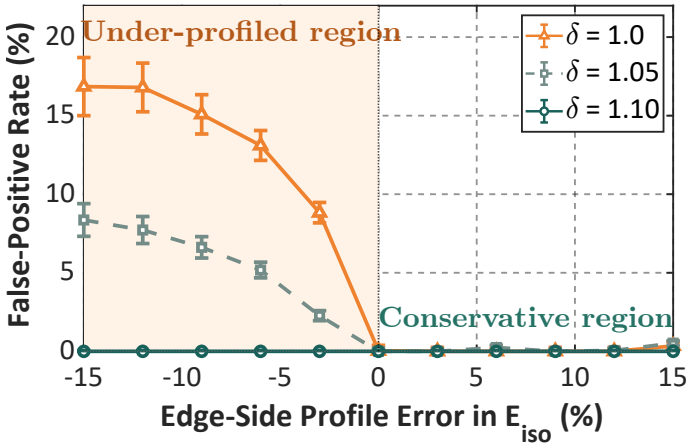
Takeaway

Hades keeps admission in hardware and wakes the MCU for admitted tasks.

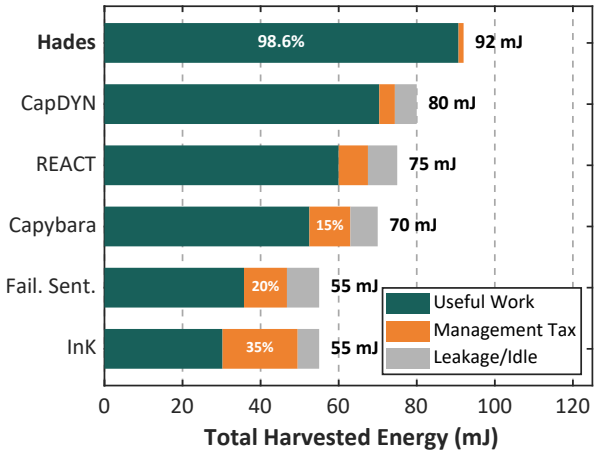
- **Energy Intake:** match sources to storage
- **Energy Aggregator:** check the combined reserve
- **Dispatch Controller:** release the right task

Let hardware decide. Let the MCU work.

Backup: threshold sensitivity to profile uncertainty



Backup: the modeled energy budget breakdown



Backup: storage topology sensitivity

